

UNIT 2: Emerging Physics for Engineering Applications

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Note: Handouts provided are supplementary materials. Students are advised to refer the prescribed text and reference books for comprehensive understanding.

EMERGING PHYSICS FOR ENGINEERING APPLICATIONS2.1 BASICS OF NANOPHYSICS :-

The idea and concept behind nanoscience started with a talk entitled "There's Plenty of Room at the Bottom" by Physicist Richard Feynman.

The study of the materials at a small scale with a condition that at least one dimension of the material should be in the range of 1 to 100 nm.

Nanophysics is a branch of Physics that focus on the study of physical phenomenon and properties of matter at nano-scale.

Nanotechnology :- The techniques involved in the preparation, characterization and use of the properties of nanomaterials in different applications are collectively called as Nanotechnology.

2.2 Significance of Nanoscale :-

When we go to nanoscale the properties of any bulk material enhanced mainly by two reasons :-

ca) Surface area to Volume Ratio :-

$$\text{For Spherical particle} \Rightarrow \frac{S}{V} = \frac{4\pi R^2}{\frac{4}{3}\pi R^3} = \frac{3}{R}$$

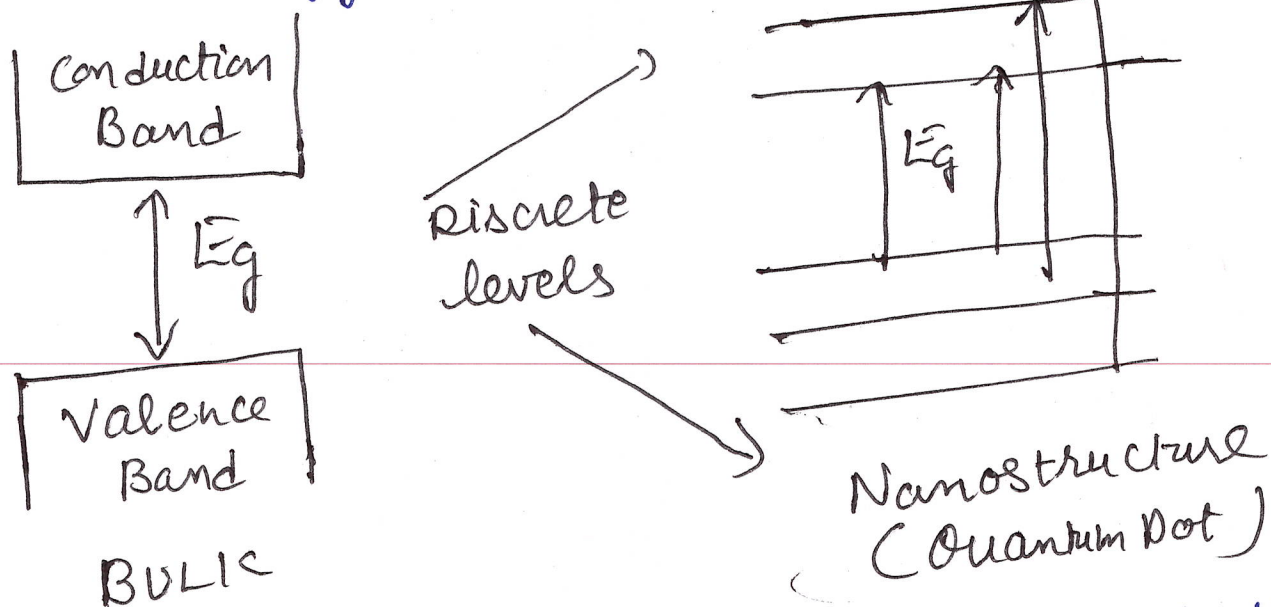
$$\frac{S}{V} \propto \frac{1}{R}$$

At surface, atoms are more reactive than in the centre.

b) Quantum Confinement :-

(2)

It occurs when the nanocrystal radius becomes comparable to the atomic Bohr radius ($0.5 \text{ \AA}$). The confinement leads to the transition from continuous to discrete energy levels.



Due to these two reasons, following properties are changed at nano-scale.

- * Opaque substances at bulk become transparent, For examples [Copper]
- * Insulator at bulk level become conductor [Silicon]
- * Gold at bulk level is solid and become liquid at nano-scale.

23.1 1D, 2D, 3D Nanomaterial :-

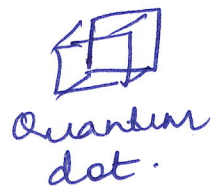
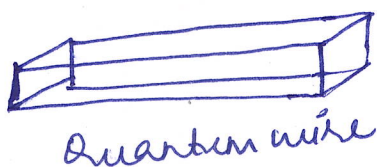
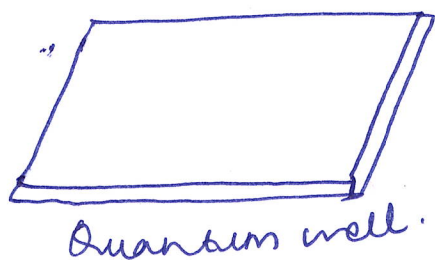
There are three dimensions to confine the bulk material. The quantum confinement needs confining at least one of these dimensions to less than 100 nanometers.

(i) If one dimension is reduced to the nano range while the other dimensions remain large, we obtain a structure known as Quantum well or 2-D structure.

(ii) If two dimensions are reduced to the nano range and one remains large, the structure is referred to as a Quantum wire or 1-D structure.

(iii) If all three dimensions are reduced to the nano range, the structure is called Quantum dot or 0-D structure.

3-D nanomaterials have no dimensions at the nanoscale. They are made up of nanomaterials. (nanowire/nanotube bundle & multiple nanolayers).



2.4. Fabrication techniques of nanomaterials:

2.4.1. Chemical Vapor Deposition (CVD)

Chemical Vapor Deposition (CVD) is a method used for the production of carbon nanotubes (CNT) and thin film deposition. It is a bottom-up approach that involves five important parameters:

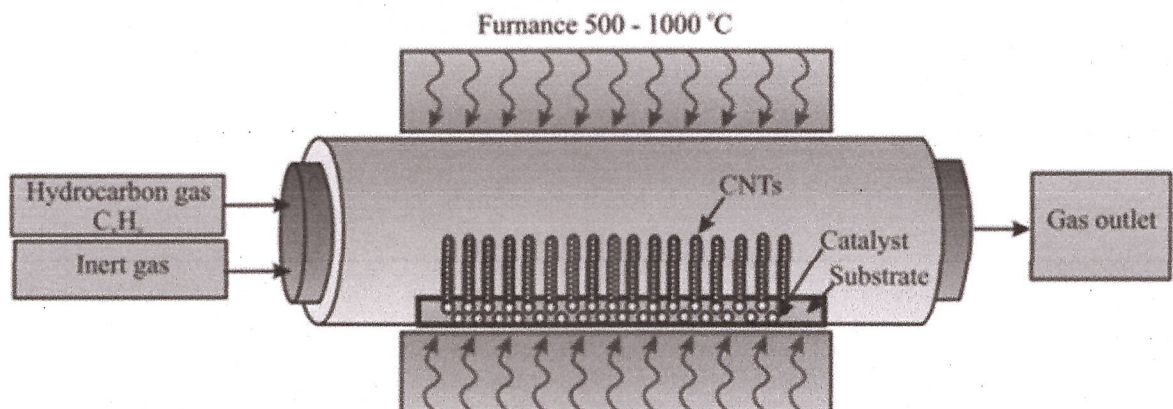
1. Carbon Source: Commonly used carbon gases include ethylene (C_2H_4) and acetylene (C_2H_2).
2. High Temperature for Reaction: The temperature is typically set at $900^\circ C$.
3. Inert Gases: Gases such as argon (Ar) and Nitrogen (N_2) are required to create an oxygen-free atmosphere.
4. Catalyst: Metals like iron (Fe) and nickel (Ni) facilitate the growth of CNTs.
5. Substrate: Quartz or Silicon serves as the surface where CNT growth begins in the furnace.

Method:

First, we start with a substrate made of silicon or quartz, onto which nanoparticles of iron (Fe) are deposited to facilitate the growth of carbon nanotubes (CNT). Argon gas (Ar) is then introduced into the furnace to eliminate any oxygen and create an oxygen-free environment. The temperature is subsequently raised to $900^\circ C$. In addition to this, the argon gas also serves as a carrier gas. Finally, a carbon-containing gas, such as ethylene (C_2H_4), is allowed to flow into the furnace.

At this elevated temperature, ethylene (C_2H_4), is absorbed onto the metal nanoparticles (Fe) over the substrate. Due to the high temperature and the inert atmosphere, hydrogen (H_2) is dissociated from C_2 and removed from the furnace, while carbon atoms deposit onto the catalyst (Fe) nanoparticles. These carbon atoms exist as single atoms—similar to free radicals—which are reactive and unstable. To achieve stability, these carbon atoms bond with one another, leading to the vertical growth of CNTs on catalyst (Fe) nanoparticles.

There is a direct correlation between the size of the catalyst (Fe) nanoparticles and the diameter of the CNTs. Therefore, using nanoparticles of controlled size allows for the production of CNTs with a specific diameter.

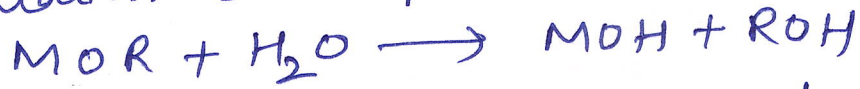


2.4.2. SOL-GEL PROCESS $\rightarrow$

This process is used for synthesis of metal oxide nanoparticles. It is also used for synthesis of composite nano powder, ceramic powder at nanoscale. Steps for preparation of nanoparticles are:

Step-1 Metal alkoxide is dissolved in water to make homogeneous solution.

Step-2 On stirring, the water molecule react with metal alkoxide (MOR) and self-formation takes place with help of hydrolysis.



Step-3 Aging of solution takes place, then sol is converted to gel. This is because some of the water molecules are evaporated. These sol particles are condensed to convert into gel (Here gel is homogeneous).

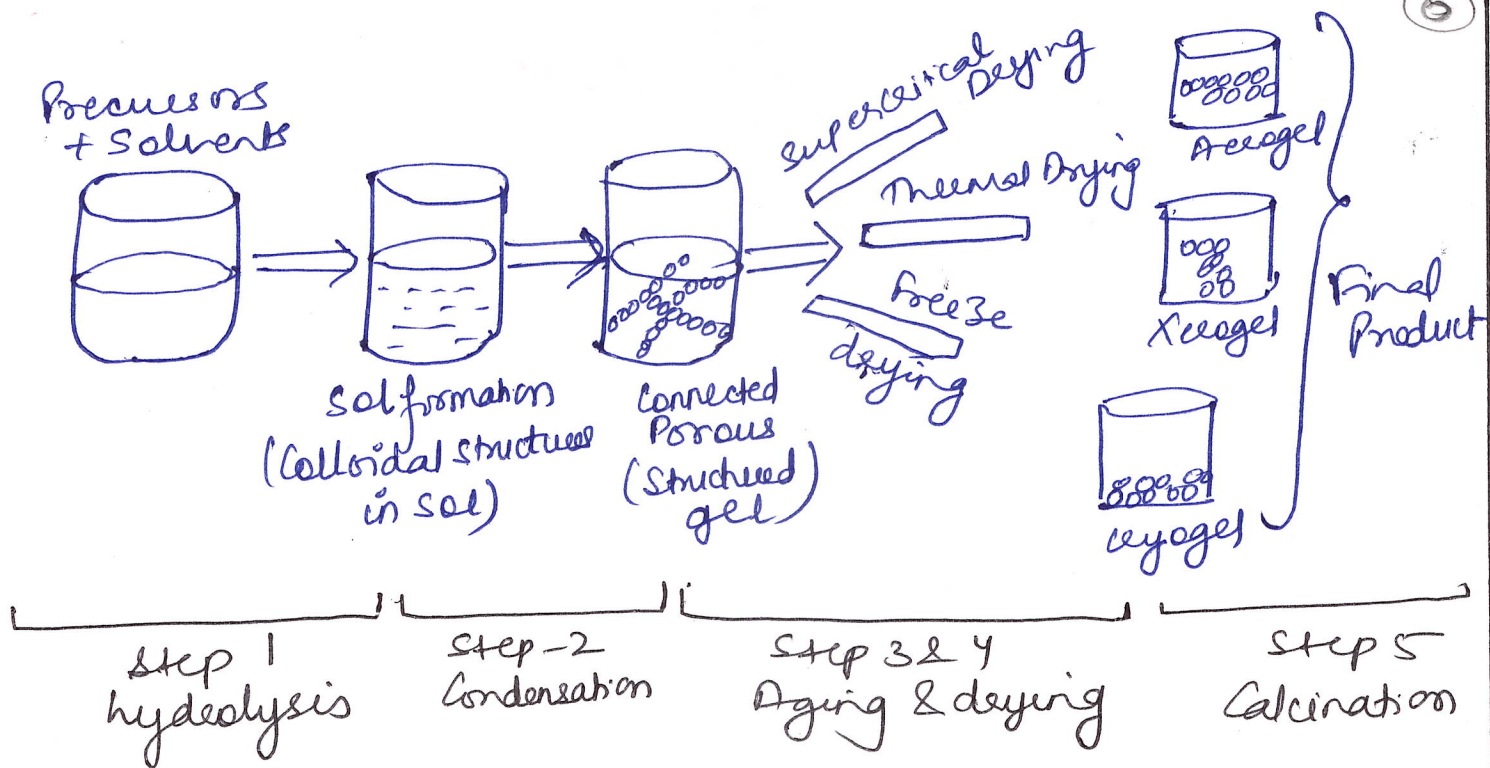


In the gel MOM bonds are formed and due to polymerization, nano materials are formed.

Step-4 To extract the nanoparticles, the solvent has to be removed by drying.

It can be done in three ways -

- (i) Super-critical drying - In this aerogel are formed by maintaining critical temperature and critical pressure.
- (ii) Thermal drying - Xerogel are formed. It can be done in room temperature.
- (iii) Freeze-drying - In this, cryogel are formed.



Step-5 Calcination - Nanoparticles gel are also formed by heating the substance with the presence of oxygen at near 800°C .

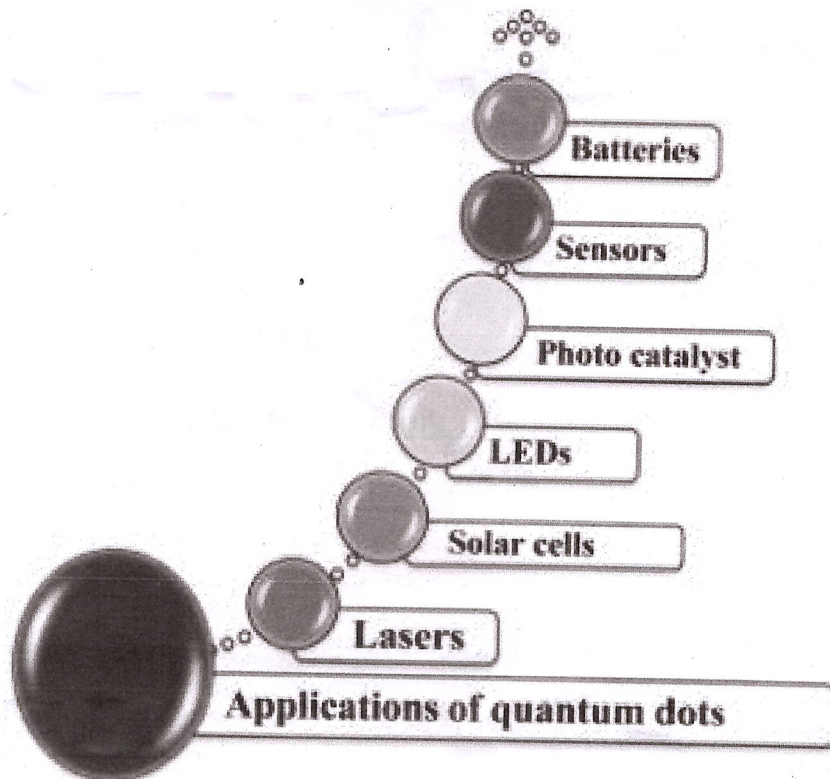
2.5. Applications of Nanotechnology

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2.5.1. Quantum Dots → Quantum Dots are used in advanced displays in (QLEDs) for vibrant colours and in solar cell to improve efficiency.

QLEDs are tiny semiconductors nanocrystals like (Indium Phosphide) having size in the range (2-10 nm) emit particular colours of light when excited. It works on the principle of recombination of electrons and holes. When an electric current is applied, QLEDs emit light through the combination of electrons and holes. The wavelength of the emitted light depends on the size and material of the quantum dots.

QLEDs can improve colour accuracy, luminous efficiency, brightness and longer life time spans.



2.5.2. Nanoscale memory: Nanoscale memory devices are significantly smaller, faster and more energy-efficient than traditional memory, transforming data storage by leveraging nanomaterials and quantum phenomena.

Various Nanoscale memory:

- Magnetoresistive RAM (MRAM): Utilizes nanometer-scale magnetic tunnel junctions, enabling instant-on computers and rapid data retrieval.
- Phase-change memory (PCM): PCM uses phase transitions to store data, providing fast write times and high endurance.
- Resistive RAM (RRAM): RRAM uses resistive switching to store data, offering high storage density and low power consumption.

Nanoscale memory Enabled by Nanotechnology

Memory Type	Core Nanotechnology	Key Materials	Advantages
Flash Memory	Nanoscale transistors	Si, Ge, advanced dielectrics	High density, low power
MRAM	Magnetic tunnel junctions (spintronics)	Magnetic nanolayers	Non-volatile, instant boot.
RRAM	Resistive oxides	TiO_2 , HfO_2	High speed, simple structure
PCM	Phase-change nanomaterials	Chalcogenides	Fast, durable non-volatile memory.
Molecule/ DNA-Band Memory	Molecular nanostructures	Organic molecules, DNA	Ultra high density, long term storage.

2.5.3. AIR & WATER Purification by nanomaterials:

Nanotechnology is leveraged in air and water purification through advanced nanomaterials that remove contaminants more efficiently than conventional methods.

Purification Area	Nanotechnology Used	Key Contaminants Removed	Mechanism/ Advantage
Water	→ Nanofiltration membranes	Heavy metals, salts, microbes	Size-selective filtration, high efficiency
	→ Carbon nanotubes, graphene	Organic/ inorganic pollutants, oil, microbes	Adsorption, antimicrobial, reusable
	→ Metal oxide nanoparticles (TiO_2 , ZnO , Ag)	Pathogens, organic pollutants	Photocatalysis, disinfection
Air	Nanoparticle-coated filters (TiO_2 , Carbon)	Particulates, VOCs (Volatile Organic compounds), Pathogens	Physical trapping, photocatalytic degradation
	Nanofibrous membranes	Ultra-fine particles, bacteria	High surface area, enhanced filtration

Nanotechnology enables removal of contaminants with higher efficiency, selectivity and frequently lower energy consumption, with broad applicability across both air and water purification systems.

2.5.4. Applications of Nanotechnology in cosmetic industry: (10)

Nanotechnology works on super tiny particles (1-100 nm) that help cosmetics work better.

→ Better skin Penetration: Nanoparticles carry active ingredients like vitamins & antioxidants deeper into skin for enhanced effects.

eg:- Anti-aging cream uses nanoliposomes.

→ Improved UV Protection: Nano-sized ZnO & TiO_2 provide strong UV protection without visible effects.

eg:- Sunscreens with nano ZnO .

→ Longer-lasting effects: Nanocapsules protect sensitive ingredients & allow slow, controlled release in creams and lotions.

eg:- Anti-wrinkle creams

Nanotechnology makes cosmetics more effective, longer-lasting, and pleasant to use by manipulating particles at the nanoscale without changing product properties.

2.6. SUPERCONDUCTORS:

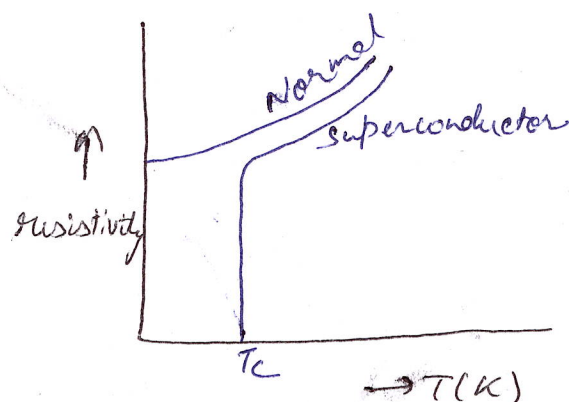
(11)

The electrical resistivity of many metals and alloys drops suddenly to zero, when the specimen is cooled upto sufficiently low temperature (temperature in liquid Helium range). This phenomenon is called superconductivity.

The material that conducts electricity with zero electrical resistance and expels magnetic field when cooled below a specific critical temperature are called superconductors.

Transition temperature: The temperature at which normal metal passes into superconducting state (resistivity disappears) is called transition temperature.

eg:- for mercury, T_c is 4.2 K



2.7. Temperature dependence of critical field: The minimum value of applied magnetic field necessary to restore resistivity (i.e. to destroy superconducting nature) is known as critical magnetic field. It is denoted by $H_c(T)$, which is a function of temperature.

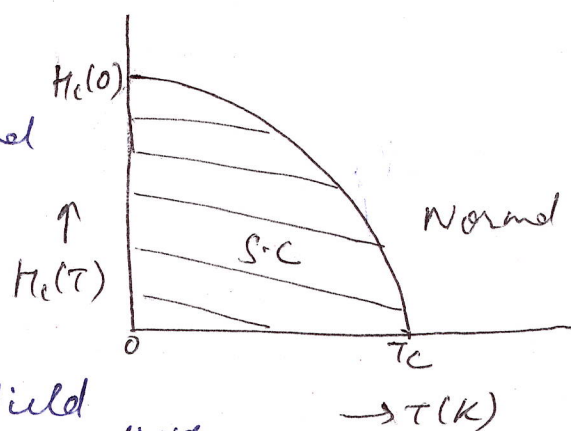
$$H_c(T) = H_c(0) \left\{ 1 - \left(\frac{T}{T_c} \right)^2 \right\}$$

where, $T = T_c$, material restores its normal state

$$\text{so, } H_c(T) = 0$$

$$\rightarrow \text{at } T = 0K$$

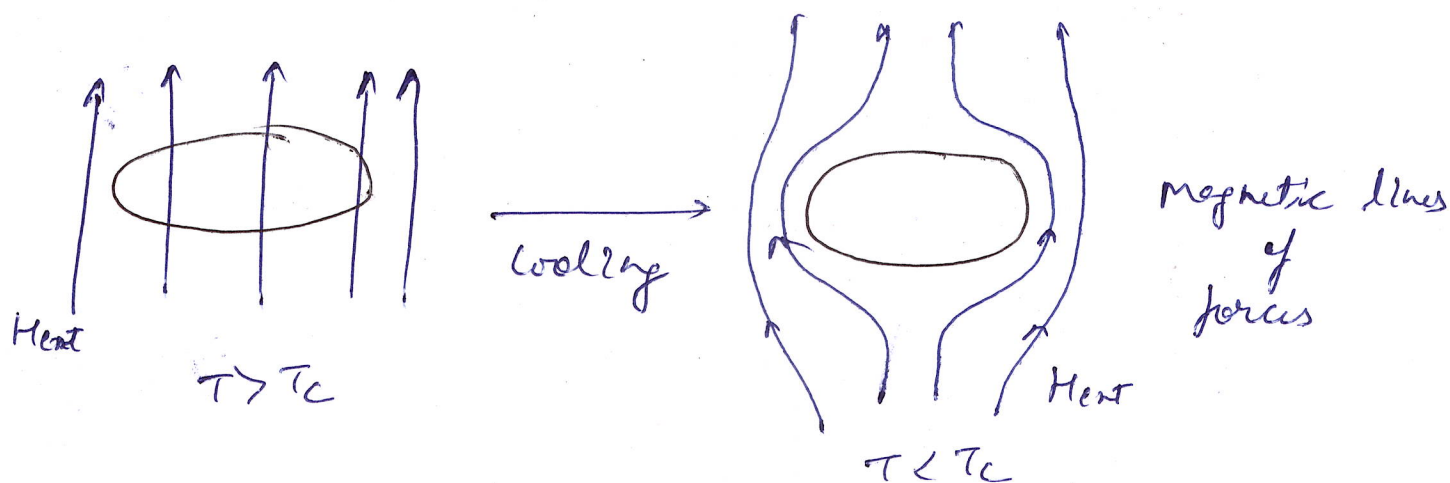
$H_c(T) = H_c(0)$, maximum critical field required to restore resistivity.



2.8. Meissner effect:

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The phenomenon of expulsion of magnetic field from the interior of superconductor when they are cooled below the transition temperature is called Meissner effect.



As $B = 0$

$$B = \mu_0 (H + M)$$

$$\Rightarrow 0 = \mu_0 (H + M)$$

$$\frac{M}{H} = -1$$

magnetic susceptibility, $\chi = \frac{M}{H} = -1$

$\Rightarrow$ Superconductor is perfectly diamagnetic.

Q. Show that zero resistivity and perfect diamagnetism are two independent and essential properties of the superconductor.

$\rightarrow$ Ohm's law, $\vec{E} = \rho \vec{J}$

As $\rho = 0 \Rightarrow E = 0$

According to Maxwell's III $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

As $\vec{E} = 0 \Rightarrow \frac{\partial \vec{B}}{\partial t} = 0 \Rightarrow \vec{B}$ is constant

Meissner effect contradicts this result.

$\therefore$ Superconductors exhibit zero resistivity and perfect diamagnetism.

CRITICAL CURRENT. $\rightarrow$

The maximum current which flows in a superconductor without losing its superconductivity. $I_c = 2\pi r H_c$

2.9. Numericals:

- a) A thin superconducting wire of a metal produces a magnetic field of $105 \times 10^3 \text{ A/m}$ on its surface due to current passing through it at a certain temperature $T \text{ K}$. Calculate T .
Given $H_c(0) = 150 \times 10^3 \text{ A/m}$ and $T_c = 9.3 \text{ K}$.

Sol) $H_c(T) = H_c(0) \left\{ 1 - \left(\frac{T}{T_c} \right)^2 \right\}$

using $H_c(T) = 105 \times 10^3$, $T_c = 9.3$ & $H_c(0) = 150 \times 10^3$
 $T = 5.03 \text{ K}$.

- Q) The transition temperature for Pb is 7.2 K . However at 5 K it loses the superconducting property subjected to a magnetic field of $3.3 \times 10^4 \text{ A/m}$. Find the maximum value of the magnetic field which allows the metal to retain its superconductivity at 0 K .

Given $T_c = 7.2 \text{ K}$ $T = 5 \text{ K}$ $H_c(T) = 3.3 \times 10^4$

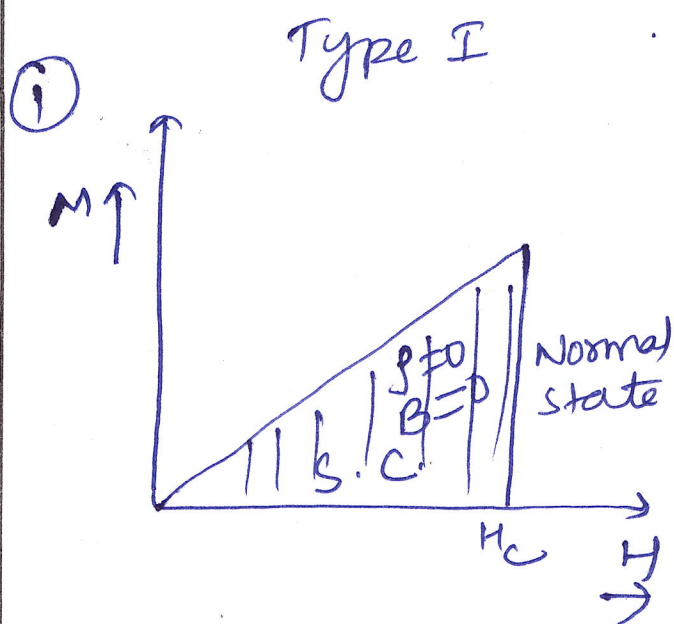
using $H_c(T) = H_c(0) \left\{ 1 - \left(\frac{T}{T_c} \right)^2 \right\}$ on solving.

$H_c(0) = 6.37 \times 10^4 \text{ A/m}$.

- Q) Determine the critical current and current density, which can flow through a long thin superconducting wire of diameter 2 mm if H_c is $1.21 \times 10^4 \text{ A/m}$.

Sol $I_c = 2\pi r H_c$ where $r = \frac{D}{2} = 10^{-3} \text{ m} \Rightarrow I_c = 75.98 \text{ Amp}$
 $J_c = \frac{I_c}{\frac{\pi D^2}{4}} = 2.42 \times 10^6 \text{ A/m}^2$

2.10. TYPE I & TYPE II SUPERCONDUCTORS. →



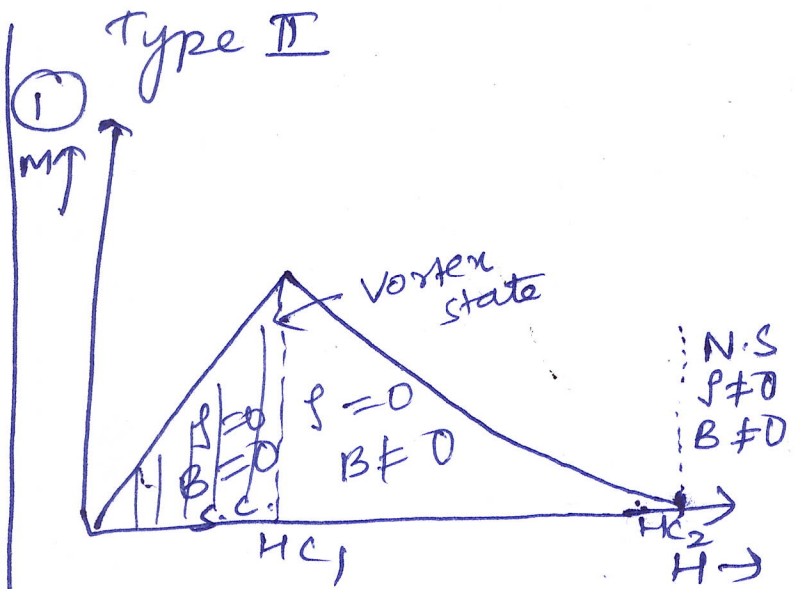
Magnetisation losses suddenly.

② Soft Superconductor

③ Exhibit Complete Meissner's effect

④ Above critical field H_c , the Superconductor converts to Normal conductor.

⑤ Examples
Pb, Sn, Hg.



Magnetisation losses gradually upto H_{c2}

② Hard Superconductor

③ Exhibit partial Meissner's effect.

④ The state between H_{c1} & H_{c2} is mixed State/Vortex state.

Above H_{c2} it converts to Normal conductor.

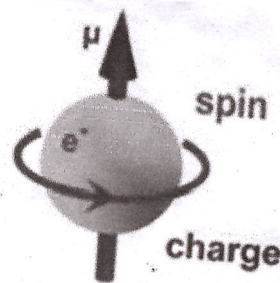
⑤ Example Nb-Sn, Nb-Ti

2.11. INTRODUCTION TO SPINTRONICS. →

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Spintronics is spin-based electronics.

Spintronics is an emerging technology that deals with both the intrinsic spin of the electron and its associated magnetic moment.

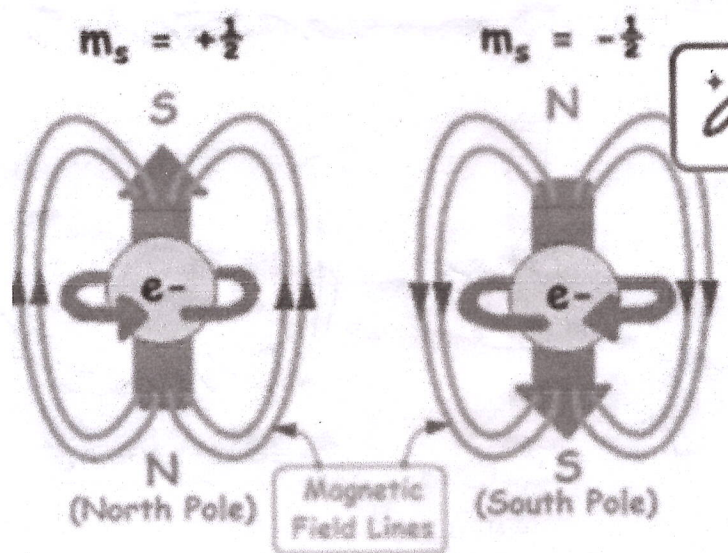


Spin is a quantum mechanical property. Here the rotational moment creates a small magnetic field.

PRINCIPLE OF SPINTRONICS. →

→ The direction and coherent motion of electron spin circulates a spin current, which will transport and store information.

→ The electron spin gives a direction up (1) and down (0).

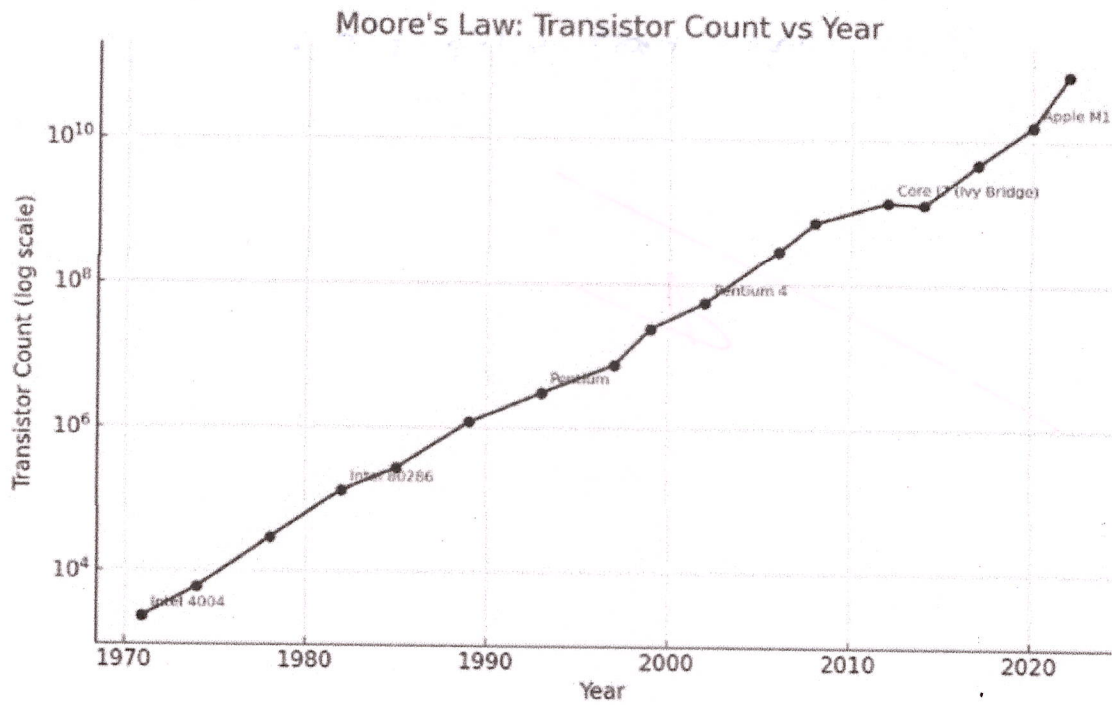


NEED OF SPINTRONICS. →

FAILURE OF MOORE'S LAW.

→ It states that the number of transistors on a silicon chip doubles approximately every eighteen months.

→ Scorching heat and consumption of high power makes the circuit inoperable.



$$\text{Transistor density} = \frac{\text{Number of transistors on chip}}{\text{chip Area}}$$

2.12. Giant Magnetoresistance (GMR) : →

The GMR effect was discovered in the late 1980. by two Scientists Peter Gruber and Albert Fert.

The GMR is a quantum mechanical effect in which electrical resistance drastically changes of a thin film made of alternating Ferromagnetic and non-magnetic layers in response to an external magnetic field.

Structure of GMR : →

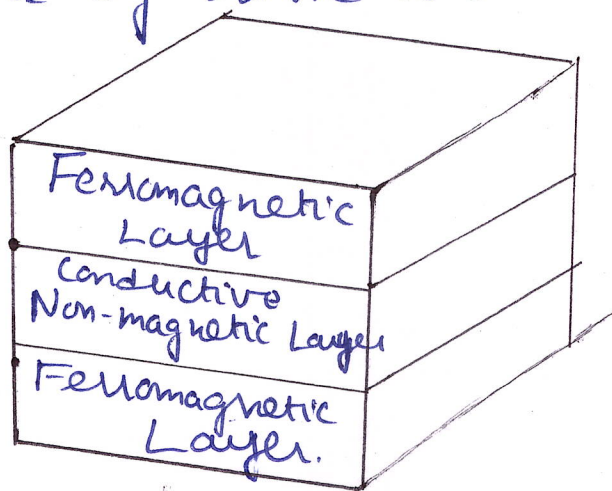
It consists of three layers : →

1.) Pinned Layer : → A Ferromagnetic Layer whose magnetization is fixed by an adjacent exchange layer.

2.) Spacer Layer : → A thin non-magnetic conductive layer sandwiched between two Ferromagnetic layers such as Copper or Chromium. The thickness lies in the nano-range.

3.) Free Layer (Sensing Layer) : → A Ferromagnetic Layer whose magnetization can change in response to an external magnetic field.

The structure of GMR is shown in fig.



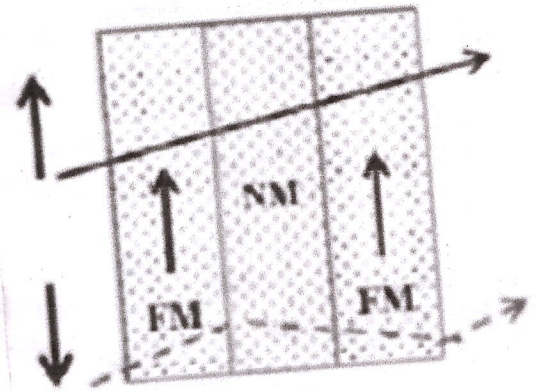
Working :- It works on the principle of electron Spin and magnetization.

A/c to the principle of Spintronics, e^- is characterized by two possible states called Spin up (majority $+\frac{1}{2}$) or Spin down (minority $-\frac{1}{2}$).

There are two types of alignment of Ferromagnetic material layers.

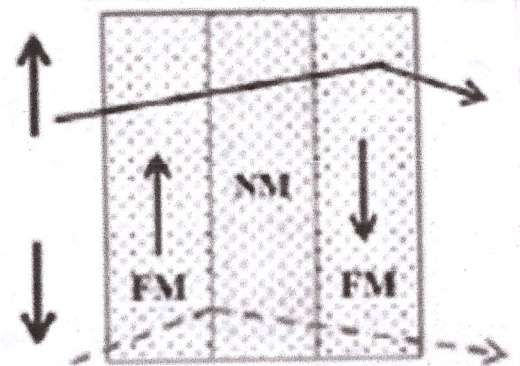
Parallel Alignment :- When an external magnetic field aligns the Free layer's magnetization in the same direction as the pinned layer, spin up e^- move through easily with no scattering and spin down e^- scatter more but conduction is dominated by majority spin.

It results in low resistance.



Anti-parallel Alignment :-

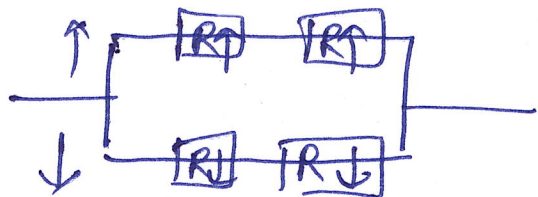
If the external magnetic field reverses the Free layer's magnetization to be opposite the pinned layer, both spin up or spin down e^- experience strong scattering in at least one layer resulting in high resistance.



Quantitative Explanation.

Case I) Parallel (FM) alignment of ferromagnetic layers

- ↑ spin electrons → weak scattering (small Resistance $R_{\uparrow}$)
- ↓ spin electrons → strong scattering (large Resistance $R_{\downarrow}$)



$R_{\uparrow}$ → Resistance due to up spin

$R_{\downarrow}$ Resistance due to down spin

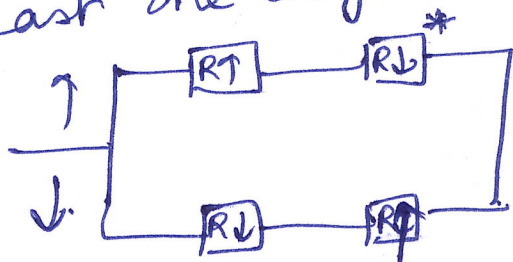
$$\Rightarrow R_{\uparrow\uparrow} = \frac{2R_{\uparrow}R_{\downarrow}}{R_{\uparrow} + R_{\downarrow}}$$

$$\approx 2R_{\uparrow}$$

as $R_{\uparrow} \ll R_{\downarrow}$
(Neglect $R_{\uparrow}$ in denominator)

⇒ Total Resistance is LOW RESISTANCE

Case II) Antiparallel (FM) alignment of layers.
→ Both ↑ & ↓ electrons scatter strongly in at least one layer.



$R_{\uparrow\downarrow}$ are in series

$$R_{\uparrow\downarrow} = \frac{R_{\uparrow} + R_{\downarrow}}{2} \approx \frac{R_{\downarrow}}{2}$$

As $R_{\downarrow}$ is Resistance due to down spin is more so neglect $R_{\uparrow} = \frac{R_{\downarrow}}{2}$

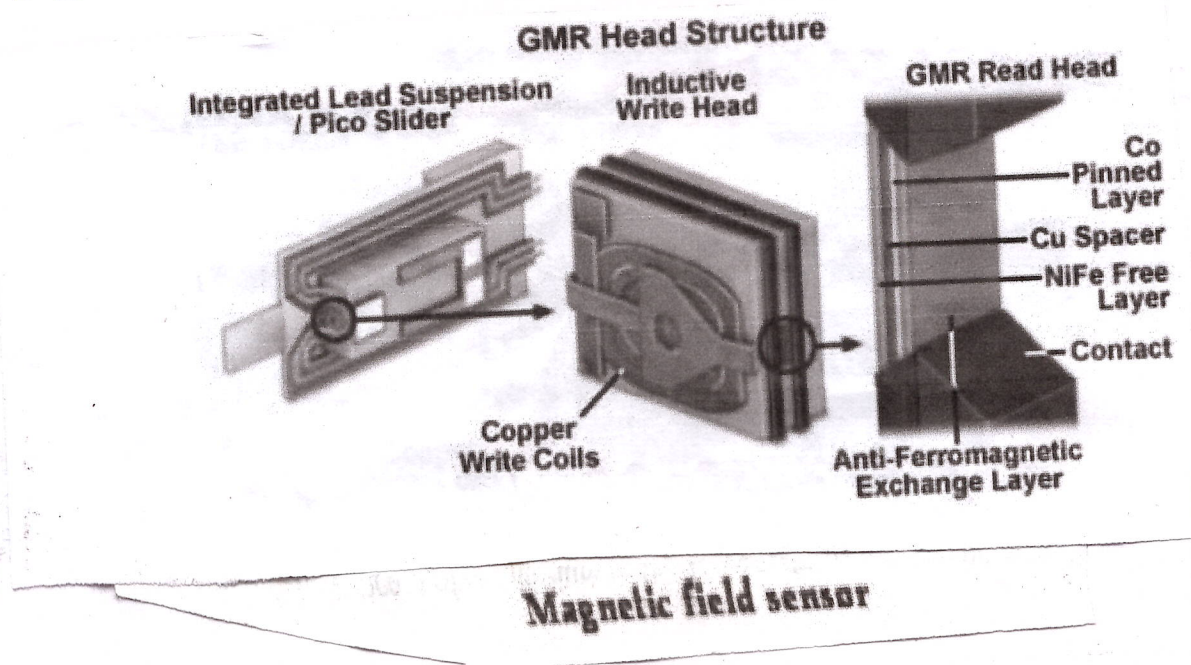
Applications :-

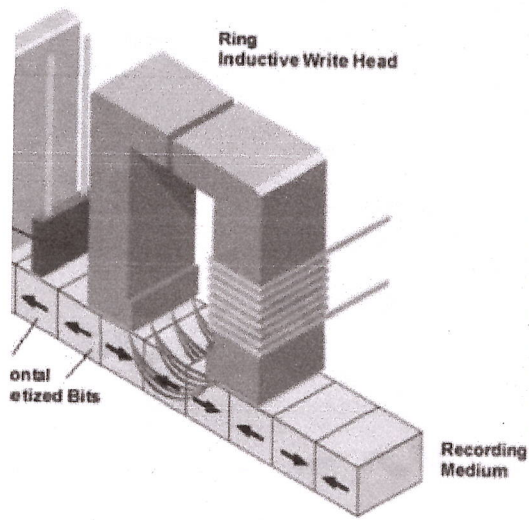
The main application of GMR is in magnetic field sensors, which are used to read data in hard disk drives, bio sensors and other devices.

GMR used in Hard disk Drives :-

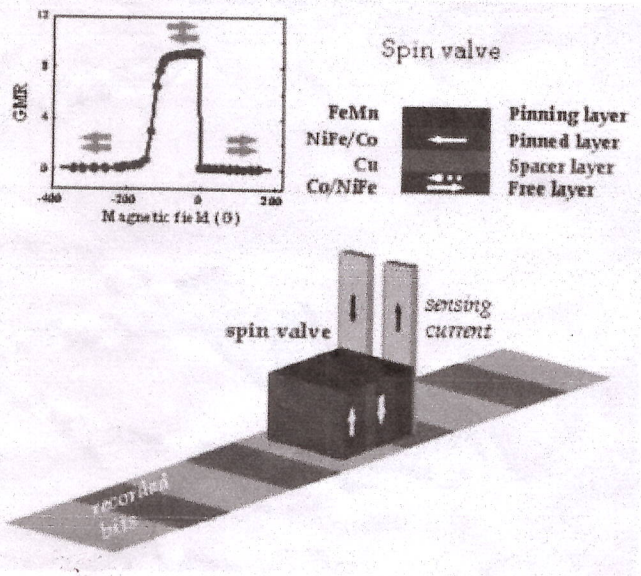
It consists of various parts :-

- 1.) GMR Read Head :- It contains a spin valve, a multilayer structure of thin films.
As the hard disk spins, the read head detects tiny magnetic data bits on hard disk platter. The magnetic field from a data bit changes the magnetic orientation of free layer.
- 2.) Resistance Change :- Magnetic field changes resistance in GMR material.
- 3.) Signal Detection :- Change in resistance detected as electric signal.
- 4.) Data Retrieval :- The signal decoded to retrieve stored data.





Magnetic field sensor



Short Questions & Answers

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Q1. Explain quantum confinement effect in nano-materials.

A. Quantum confinement occurs when nanocrystal radius becomes comparable to the atomic Bohr radius. The confinement leads to the transition from continuous to discrete energy levels.

Q2. Write two applications of nanomaterials.

A. Nanomaterials are used in nanoscale memory and air & water purification process.

Q3. Define transition temperature.

A. The temperature (T_c) at which some specific material loses its resistivity is known as transition temperature.

eg:- T_c for Hg is 4.2 K.

Q4. What is the value of critical field of a superconductor at transition temperature?

A. The value of critical field is zero at transition temperature.

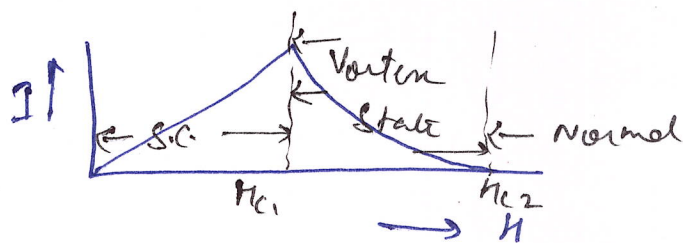
$$H_c(T) = H_c(0) \left\{ 1 - \left(\frac{T}{T_c} \right)^2 \right\}$$

at $T = T_c$

$$H_c(T) = 0$$

Q5. What is vortex state of superconductivity?

A. Vortex state is an intermediate state in between H_{c1} & H_{c2} of Type II superconductors, where the specimen exhibit partial Meissner effect.



Q.6. Which quantum property is central to spintronics?

A. The spin of the electron is the central quantum property in spintronics.

Q.7. What is the main principle of spintronics?

A. The direction and coherent motion of electron spin create a spin current, which is used to transport and store information.

Q.8. What does the rotational motion of electron spin create?

A. It creates a small magnetic field and magnetic moment.

Q.9. What is the main application of AMR?

A. The main application of AMR is in magnetic field sensors used to read data in hard disk drives, biosensors, and other devices.

Q.10. On what principle does AMR technology work?

A. AMR works on the principle of electron spin and magnetization (spintronics), where electron conduction depends on spin alignment in magnetic layers.

Practice Questions

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- Q1. What are the changes in the properties that takes place in a material when its size reduce to nano scale.
- Q2. What are nano materials? Explain briefly the basic concepts of Quantum Dots, Quantum wires and Quantum well.
- Q3. What is the purpose of nanoscience? Discuss any one method (CVD/Sol-gel) for the synthesis of nanomaterials.
- Q4. Discuss the applications of nanomaterials in the field of Quantum Dots, nanoscale memory and air & water purification.
- Q5. What do you mean by superconductivity? What is transition temperature?
- Q6. Explain Meissner effect.
- Q7. Explain Type I & Type II superconductors
- Q8. A superconducting material has critical temperature of 4.2 K in zero magnetic field and a critical field of 0.0306 T at 0 K . Find the critical field at 2.1 K .
- Q9. For a specimen of V_3Ga , the critical fields are respectively 0.176 T & 0.528 T for 14 K & 13 K . Calculate the transition temperature and critical fields at 0 K & 4.2 K .
- Q10. Calculate the critical current which can flow through a long thin superconducting wire of diameter 10^{-3} m , given, $H_c = 7.9 \times 10^3\text{ A/m}$.

- Q.11) What is spintronics? What is the need of spintronics?
- Q.12) Explain the construction and working of Giant Magnetoresistance with a suitable diagram.
- Q.13) Explain parallel alignment and antiparallel alignment on the basis of spin.
- Q.14) How Quantum dot plays a major role in nanotechnology?
- Q.15) Explain in brief nanoscale memory, air and water purification and cosmetic industry as the applications of nanotechnology.

Summary

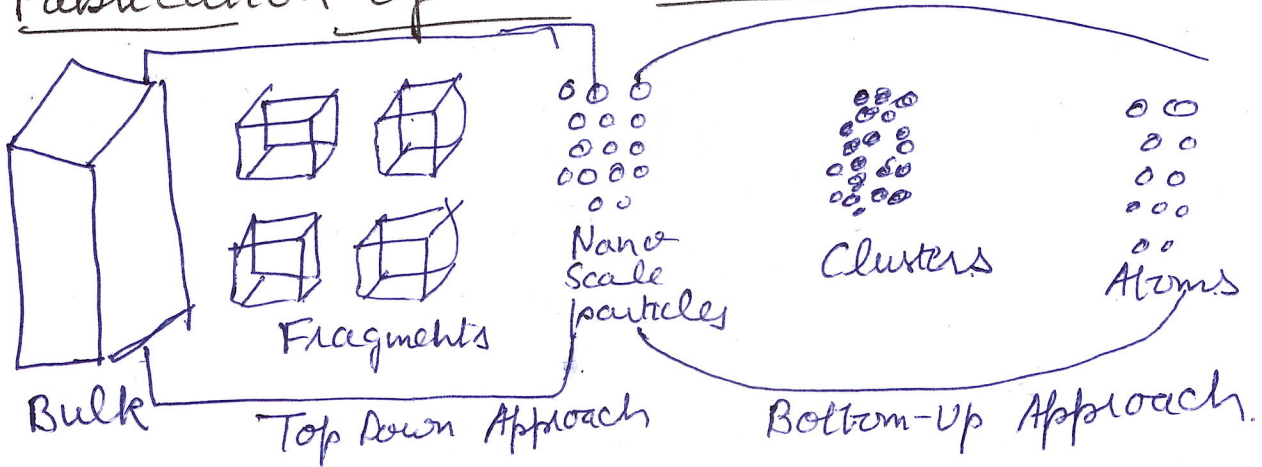
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Nanotechnology

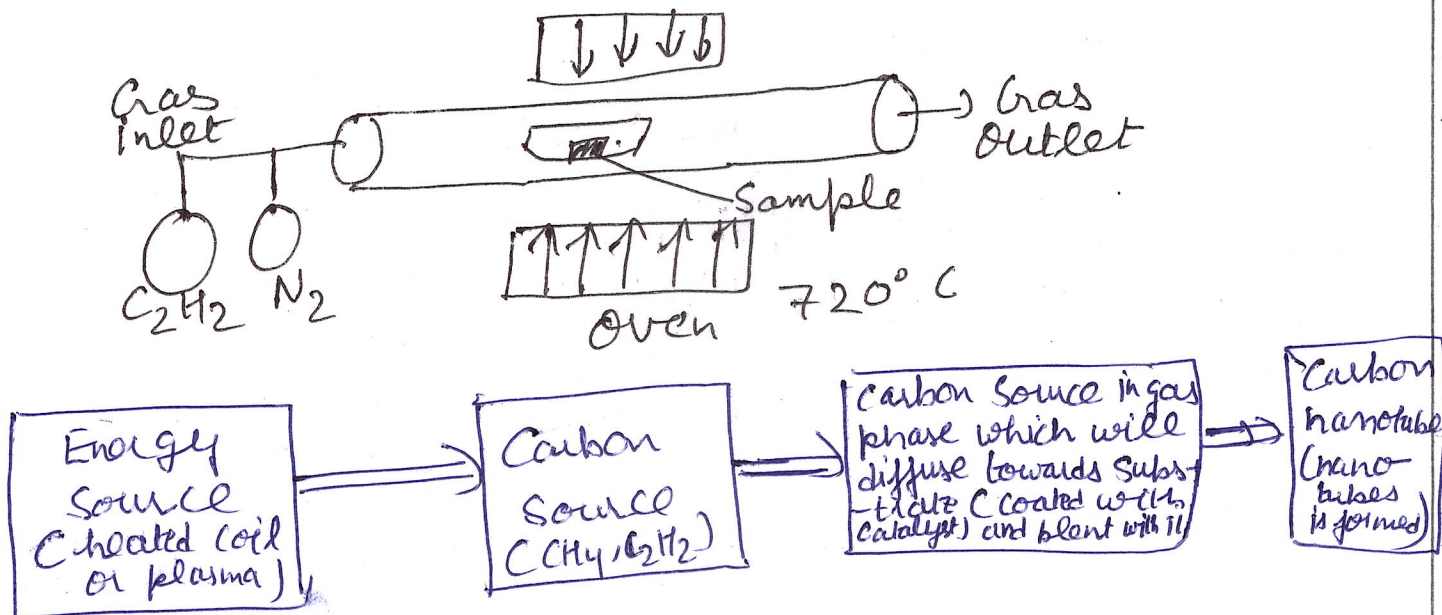
* Nanomaterials have particle dimension in nano-range (1-100nm).

* 1 Dimensional nanomaterial	2 Dimensional nanomaterial	3-Dimensional nanomaterial
When one dimension is reduced to nano-range. This structure is called Quantum well or 1D.	When 2-dimensions are reduced to nano-range. This structure is called Quantum wire or 2D.	When 3-dimensions are reduced to nano-range. This structure is called Quantum dot or 3D.

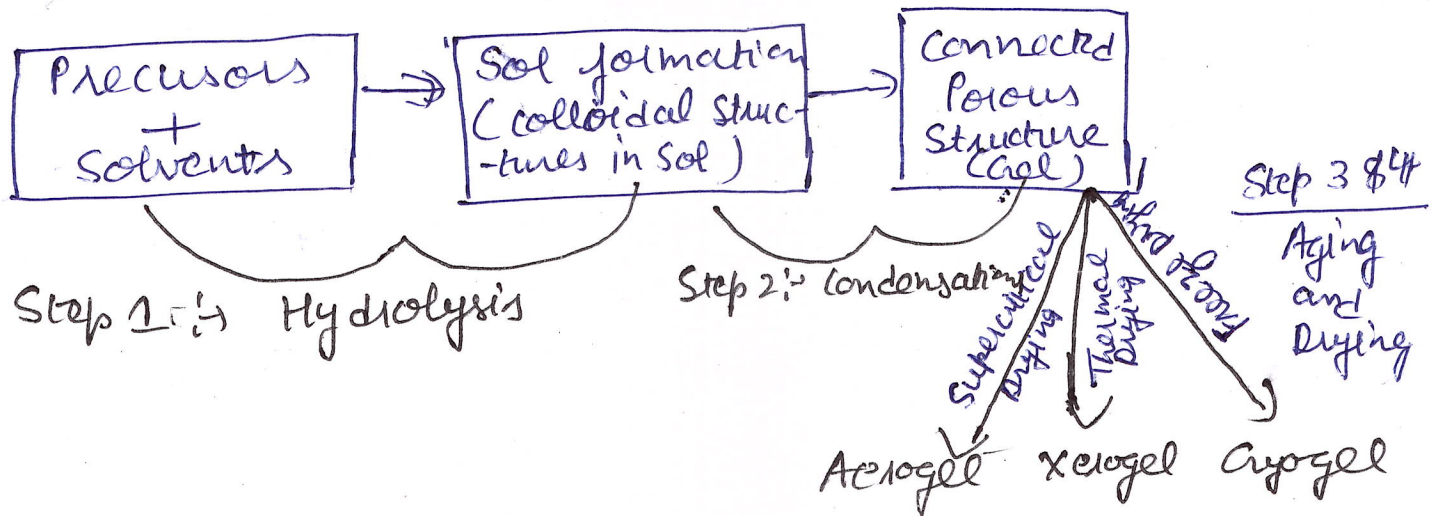
* Fabrication of Nano-materials :-



* Chemical Vapour Deposition (CVD) method :-



* Sol-Gel Method :-

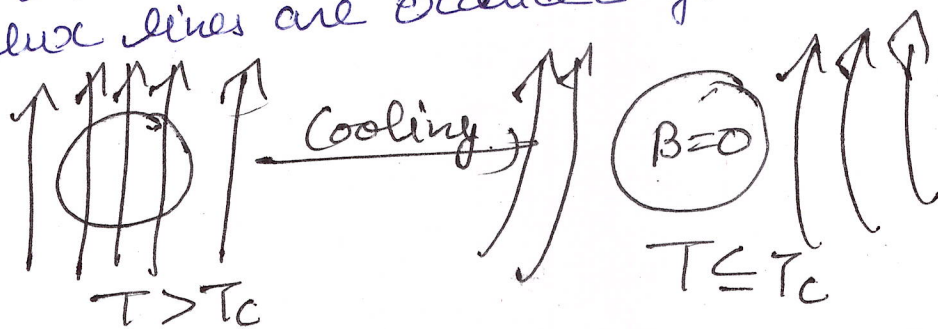


* Applications of Nanotechnology :-

- Quantum dots are used in displays that are cheaper, lighter, brighter and more efficient.
- They are used in renewable energy ultra high performance solar cells.
- Nanoparticles are used in antibacterial silver coating on wound dressing.

* Meissner Effect :-

When superconducting material is placed in a magnetic field under the condition when $T \leq T_c$ and $H \leq H_c(T)$, the flux lines are excluded from the material,



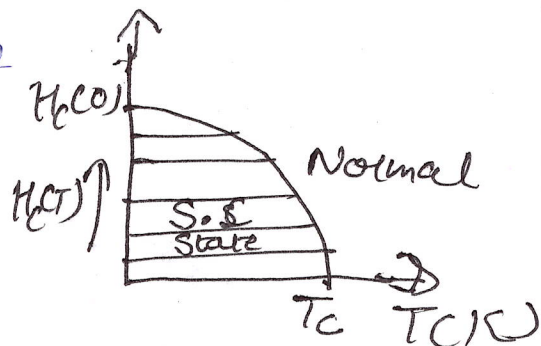
* Critical Magnetic field :- The minimum magnetic field required to destroy the superconducting property.

$$H_c(T) = H_c(0) \left\{ 1 - \left(\frac{T}{T_c} \right)^2 \right\}$$

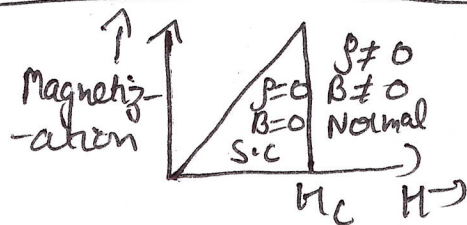
$H_c(0)$ → Critical field at 0K

T_c → Critical Temperature

$H_c(T)$ → Critical field at temperature T

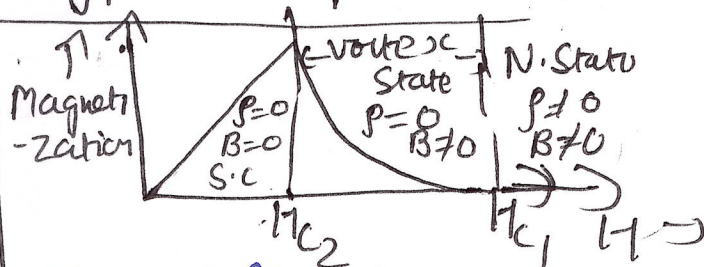


* Type-I Superconductors



- They exhibit perfect Meissner effect below H_c .
- Ex - Pure metals

Type-II Superconductors



- They follow partially Meissner effect between H_{c1} and H_{c2}
- Ex - Alloys.

Spintronics :-

Spintronics is an emerging technology that utilizes the intrinsic spin of electrons, in addition to their charge to store and manipulate information leading to faster, smaller and more energy-efficient devices.

- * The basic principle of spintronics is the direction and coherent motion of e^- . The e^- spin gives a direction up (1) and down (0).

The applications in magnetic readers for hard drives (GMR).

Giant Magnetoresistance (GMR) :- GMR

- works on the principle of electron spin and magnetization.
- It is quantum mechanical effect in which electrical resistance drastically changes due to thin film made of alternating ferromagnetic & non-magnetic layers in response to an external magnetic field.
- GMR is used in hard disk drives to store data.